

Estimation of transport load capacity with complex surface geometry cargo

In collaboration with
Company Name



Student(s)
Student Name, email, phone number
Student Name, email, phone number
Student Name, email, phone number
Student Name, email, phone number

Program
Program Title

Skoltech supervisor
Skoltech supervisor Name

Company supervisor
Company supervisor Name

Introduction

The aim of this project was to create software for accurate measurement of cargo volume with the intention of integrating it into production processes to improve the quality of work. These measures are necessary to improve the operational efficiency of the company, as well as the durability of the equipment.
By implementing an automated system using Lidar technology, we sought to improve accuracy, reduce human error and optimize workflows.

Objectives

- To develop a reliable system for accurate measurement of cargo volume.
- Implement an automated solution to optimize the workflow.
- To test the created system in real time.
- Ensure smooth integration of the new system with the company's existing management systems.

Process & Methods

Installation development

The design of the installation was developed and agreed upon (Fig.1). The installation consists of:

1. An aluminum housing
2. A source of constant lighting
3. Two 2D LIDARs positioned perpendicular to each other
4. RGB camera, which is used to record and check a passing vehicle
5. Radio tags for reading the model of the approaching vehicle and activating the software.

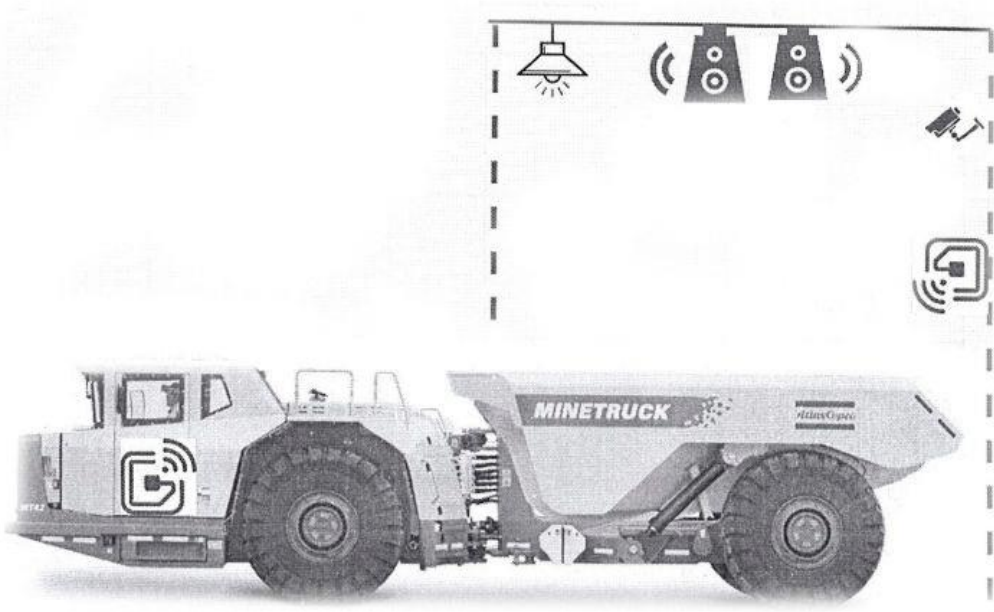


Figure 1: An illustration of equipment installation

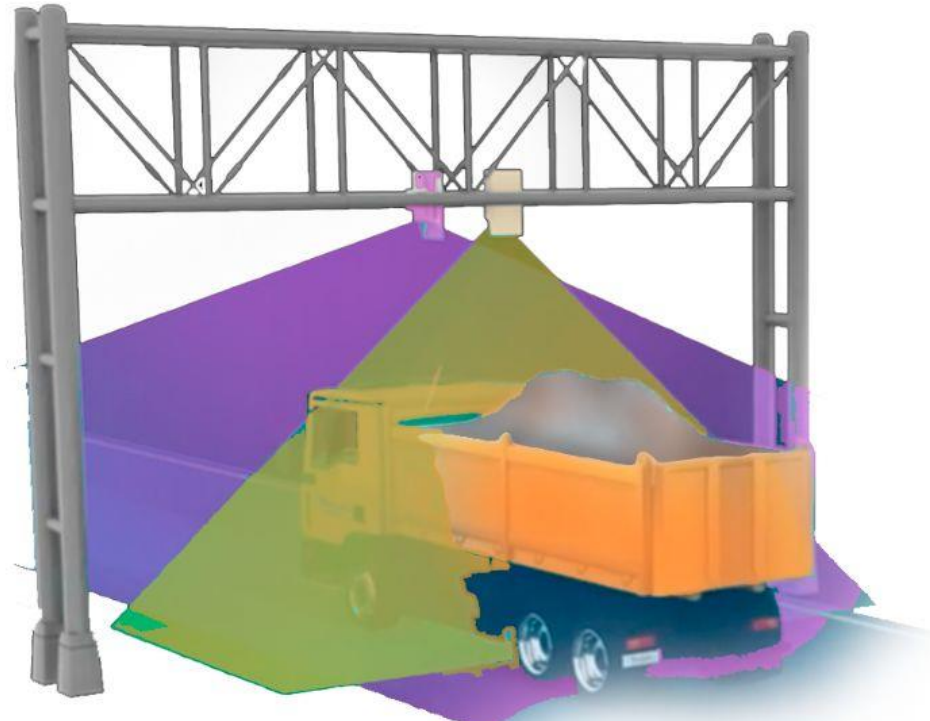


Figure 2: An illustration of a cargo scan

Volume calculation system

Utilizing ROS 2 and Opened, a system for determining the volume of cargo was developed (Fig. 3) that can easily integrate after scanning a moving vehicle (Fig. 2).

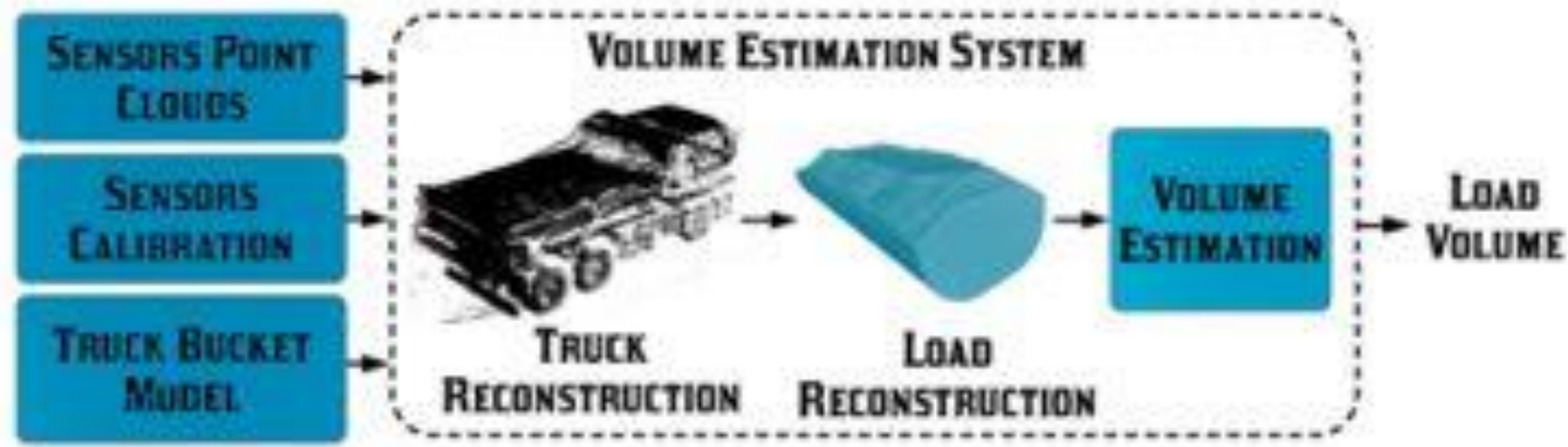


Figure 3: Overview of the system for estimating the load volume

The algorithm consists in combining two point clouds to obtain a three-dimensional point cloud after determining the route model and calibrating the sensors. After that, the point cloud corresponding to the real vehicle is restored in volume. Then, using segmentation, the points related to the cargo are determined and a three-dimensional model of the cargo is created. Next, the volume of the resulting shape is calculated. Finally, the actual volume of the cargo is determined. This algorithm of operation allows you to accurately and quickly get an accurate result.

Results

The developed system accurately determines the cargo volume in trucks used in blasting operations through the use of dual 2D LiDARs and advanced data processing algorithms.

The system integrates seamlessly with existing production management frameworks, providing real-time volume data that enhances inventory management and logistics planning. The following key results were achieved:

- **Accurate Volume Measurement:** The system reliably measures cargo volume with high precision, minimizing errors associated with manual methods.
- **Real-time Processing:** Automated data collection and processing ensure continuous, real-time updates of cargo volume.
- **Seamless Integration:** The system's compatibility with existing production systems allows for immediate use and operational efficiency.
- **Enhanced Inventory Management:** Real-time volume data supports better resource allocation and logistical planning.

The system was thoroughly tested and validated, demonstrating its effectiveness and reliability in various operational conditions.

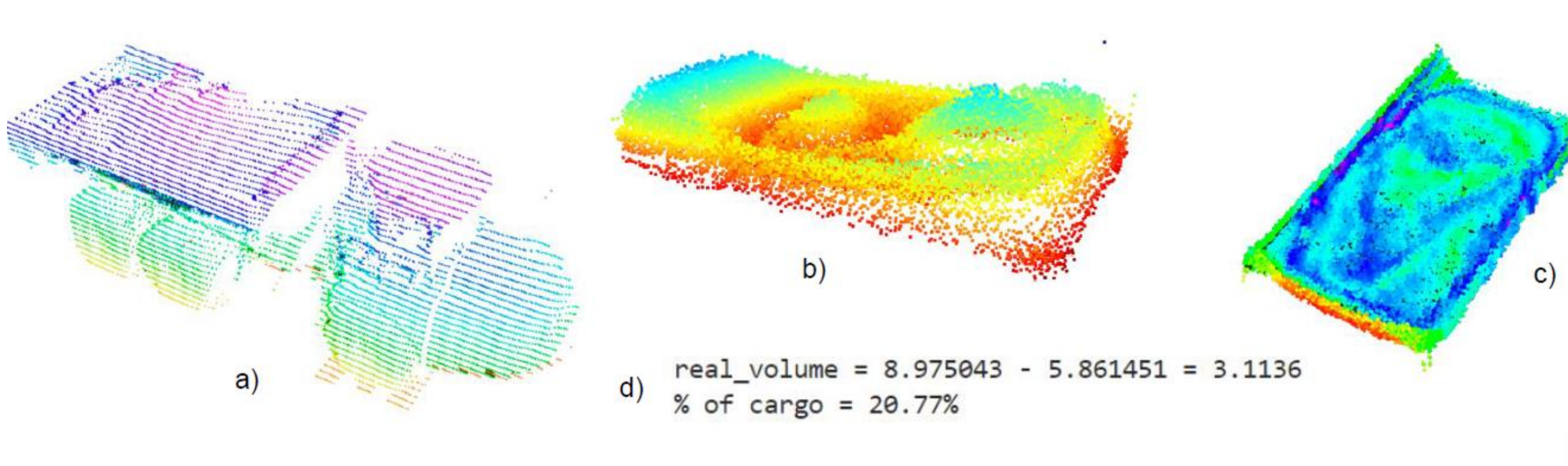


Figure 4: Illustration of the real work process

- Figures 4a-d illustrate the system's capabilities:
- Figure 4a: Point cloud of the entire truck, showing the comprehensive scan of the vehicle and cargo.
 - Figures 4b-c: Segmentation of the cargo from the point cloud, highlighting the isolation of the load from the rest of the truck.
 - Figure 4d: Result of data processing using the developed program, demonstrating the final calculation and visualization of the cargo volume.

Conclusions

This project presents an approach to cargo volume measurement in trucks by employing dual 2D LiDARs and advanced data processing techniques. The developed system enhances accuracy and efficiency through automated, real-time scanning and volume estimation. It integrates seamlessly with existing production management systems, improving inventory management and logistics planning. The solution has significant potential applications in various industrial settings where precise cargo volume measurement is critical. Future areas of work include the integration of this work in real time in real production.